HINTS AND SOLUTIONS

Q.1 C
Q.2 A
Q.3 B
Q.4 D
Q.5 A
Q.6 B
Q.7 A
Q.8 B
Q.9 C
Q.10 C

Sol. Velocity flowing per sec $= Av = A\sqrt{2gh}$

$t.Av = Ah$

or, $t = \sqrt{\dfrac{h}{2g}}$

Now, $A\sqrt{\eta^2 gh} \times t' = \eta Ah$

$t' = \sqrt{n}\dfrac{\sqrt{h}}{\sqrt{2g}} = \sqrt{\eta\dfrac{h}{2g}} = \sqrt{\eta t}$

Q.11 A

Sol. F_b is same for shell and solid sphere

$W - F_b = \dfrac{W}{g}a$ for hollow shell

$W - F_b = \dfrac{W'}{g}a'$ (for solid shell)

$W\left(1-\dfrac{a}{g}\right) = W'\left(1-\dfrac{a'}{g}\right)$

As $W' > W$

$a < a'$

Q.12 A
Q.13 AC

Sol.$_{NB}$ Half immersed = 50 cc displaced water. So 25cc rise in left and 25cc rise in right according to archemedies principle, buoyant force is equal to weight of displaced fluid, here, is equal to weight of 50cc water and if is less than the w.r.t. of 100cc ice ball, so, the ball cannot float.

Q.14 AB

Sol. (A) If ice melts then its excess volume outside the water will raise the water volume i.e., 50cc is displaced but 90cc will be the increase in volume of water by melting 100cc ice.

(B) If water freez then it happens around the central crystal so volume of ice ball will increase and it will take in surrounding water so, water level falls on both sides.

Q.15 AC

Sol. (A) Heating will melt the ie and increase the volume of water.

(C) coin placed in right vessel will displace some volume of water, which will be equally distributed to both vessels raising level in both.

Q.16 BD

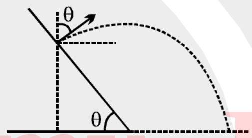
Sol. In both cases $V_A = V_3$

$$\frac{p_A}{p_g} + \frac{v^2}{y} = \frac{p_B}{p_g} + \frac{v^2}{y} + h$$

$p_A > p_B$ in both cases.

Q.17 AD

Sol. $v = \sqrt{2g(\Delta h)} = \sqrt{2\times10\times(5/100)} = 1$ m/s



Range $= 0.8 \times \frac{1}{10} = 8$ cm

$\therefore$ Range from B $= 8$ cm $- 8$ cm $\times \frac{3}{5} = 3.2$ cm

$$v\cos\theta\, t - \frac{1}{2}gt^2 = -0.08$$

$$0.6t - 5t^2 = -0.08$$

$$t = \frac{0.6 \pm \sqrt{0.36 + 4\times5\times0.08}}{2\times10} = \frac{0.6\pm1.4}{20} = \frac{1}{10}$$

Horizontal force required $= mv\cos 37° = \rho v^2 \cos 37° = 1.5 \times 10^{-4} \times 1000 \times 2 \times g \times \Delta h \times (4/5)$

$$= 1.5 \times 2 \times \frac{0.05}{100} \times \frac{4}{5} = 0.12 \text{ N}$$

Q.18 CD

Q.19 ABD

Q.20 [Ans H =3m]

Sol. At the "critical" depth H, the equilibrium condition for the compound object suggests that the density of the liquid is $d = (d_1 + d_2)/2$. For the top and the bottom object taken separately, the equilibrium conditions are, respectively:

$d_1Vg + F - (dVg - dgAH) = 0$

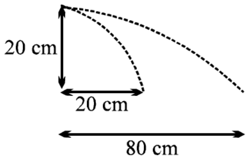
$d_2Vg - F - (dVg + dgAH) = 0$

(For each object, the term in parentheses indicates the " effective" buoyancy force.)

Combining the last two equations yields the answer : $H = [(d_2 - d_1)Vg - 2F] / [(d_1 + d_2)\, gA]$

Q.21 [Ans.(a) 80cm, 5cm; (b) 300sec.]

Sol.$_{nv}$(a)

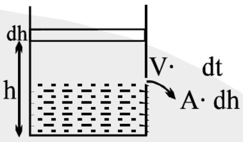


$\Rightarrow \quad 20 \times 10^{-2} = v_1 \cdot \sqrt{\dfrac{2 \times 20 \times 10^{-2}}{10}} \quad \Rightarrow \quad v_1 = 1 \text{ m/s}$

$\rho g h_1 = \dfrac{1}{2}\rho \cdot v_1^2 \quad \Rightarrow \quad h_1 = 5 \text{ cm}$

$\Rightarrow \quad 80 \times 10^{-2} = v_2 \sqrt{\dfrac{2 \times 20 \times 10^{-2}}{10}} \quad \Rightarrow \quad v_2 = 4 \text{ m/s}$

$\Rightarrow \quad \rho g h_2 = \dfrac{1}{2}\rho \cdot v_2^2 \quad \Rightarrow \quad h_2 = 80 \text{ cm}$



(b) Volume coming out from the orifice in dt time = A · dh

$\therefore A \cdot dh = -\dfrac{V \cdot A \cdot dt}{1000}$

$v = \sqrt{2gh} \times \left\{\rho h g = \dfrac{1}{2}\rho V^2\right\} \quad \Rightarrow \quad \dfrac{dh}{V} = -dt$

$\Rightarrow \quad \displaystyle\int_{80}^{5} \frac{dh}{\sqrt{2gh}} = \int_0^t -dt$

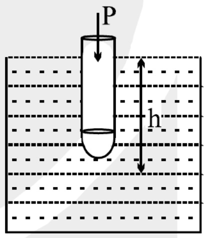
$\therefore$ time = 300 sec

Q.22 [Ans. (a) 1.01146×10^5 (b) 14.6 mm]

Sol. Excess pressure = 2s/r = 146 Pa

Pressure 'h' below the surface = $\rho_0 + \rho g h$

$\therefore \quad P_{\text{inside bubble}} = (P_0 + \rho g h) + 2s/r$



$= 1 \times 10^5 + 10^3 \times 10 \times 0.1 + 2 \times \dfrac{7.3 \times 10^{-2}}{10^{-3}}$

$= 1 \times 10^5 + 10^3 + 146$

$= 1 \times 10^5 + 1146$

$= 1.01146 \times 10^5$

(b) $h = \dfrac{2s}{\rho g r} = \dfrac{2 \times 7.3 \times 10^{-2}}{10^3 \times 10 \times 10^{-3}} = 14.6 \times 10^{-3} \text{ m} = 14.6 \text{ mm}]$